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PROVISIONAL PATENT APPLICATION UNDER 35 U.S.C. § 111(b)

APPARATUS, SYSTEMS, AND METHODS FOR NATIVE 16-DIMENSIONAL SEDENION SIGNAL PROCESSING, PRE-ADC ANALOG ZERO-DIVISOR INTERFERENCE CANCELLATION, SINGLE-SNAPSHOT NON-ASSOCIATIVE CURVATURE SPOOFING DETECTION, AND ZERO-DIVISOR KEYED MODULATION

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CROSS-REFERENCE TO RELATED DISCLOSURES (35 U.S.C. § 102(b)(1) & MPEP § 2153)

[0001] This Provisional Patent Application is filed under 35 U.S.C. § 111(b), 37 CFR 1.53(c), and MPEP § 201.04, and claims the benefit of the one-year statutory grace period under 35 U.S.C. § 102(b)(1)(A) (MPEP § 2153.01(a), United States) and Article 12 of Law No. 9.279/96 (Brazil INPI) with respect to the inventors' own technical disclosures: (1) Pedro Barone Agourakis and Dionisio Chiuratto Agourakis, "The Sedenion Triangular Root: A Child-Originated Partial Desimplicialization Operator on Cayley–Dickson Bitmasks," Zenodo DOI 10.5281/zenodo.21246673 (July 7, 2026); (2) Pedro Barone Agourakis and Dionisio Chiuratto Agourakis, "Zero-Divisor-Free Coordinate Cells in the Cayley–Dickson Tower," Preprint v1.0 (July 29, 2026); (3) Dionisio Chiuratto Agourakis, "Sedenion Algebras and PG(3,2) Incidence Structures for Non-Associative Geometric Deep Learning: Algebraic Foundations, Machine-Checked Theorems, and Dual-Track Validation," Zenodo DOI 10.5281/zenodo.23588274 (October 2026); and (4) Dionisio Chiuratto Agourakis, "Matrix-Inversion-Free O(1) Sedenion PG(3,2) Zero-Divisor Beamforming for Assured PNT, +80 dB GNSS Anti-Jamming, and Anti-Spoofing," Zenodo DOI 10.5281/zenodo.23603006 (October 2026). All four disclosures are incorporated herein by reference in their entirety pursuant to 37 CFR 1.57 and MPEP § 608.01(p).

FIELD OF THE INVENTION (MPEP § 608.01(c))

[0002] The present invention relates generally to physical electromagnetic array receivers, passive radio-frequency (RF) stripline directional-coupler networks, pre-ADC analog differential interference cancellation circuits, satellite navigation (GNSS) anti-jamming and anti-spoofing hardware, and low-probability-of-intercept / low-probability-of-detection (LPI/LPD) physical-layer transceivers. More specifically, the invention relates to passive reciprocal 32-port RF coupler matrices, pre-ADC switched analog differential crossbar networks, single-snapshot non-associative curvature detectors, and zero-DSP48 digital processors operating natively in the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S} \cong \mathbb{O} \oplus \mathbb{O}e_8$ and the finite projective geometry $\mathbf{PG}(3,2)$.

BACKGROUND OF THE INVENTION

[0003] All conventional multi-antenna digital signal processing (DSP), phased-array radar, and Controlled Reception Pattern Antenna (CRPA) receivers formulate received wavefields as vectors in an N-dimensional complex Hilbert space $\mathbb{C}^N$ governed by associative, commutative complex scalar multiplication. While foundational, the $\mathbb{C}^N$ linear-algebraic paradigm suffers from four fundamental physical and computational limitations in contested electromagnetic environments:

[0004] 1. Pre-Beamformer ADC Saturation, LNA Compression, and Co-Linear Blindness: In conventional Digital Beamforming (DBF), raw RF/IF signals from N antenna elements are amplified by Low-Noise Amplifiers (LNAs) and digitized by Analog-to-Digital Converters (ADCs) prior to spatial null-steering. Under high-power Electronic Attack (EA) where the Jammer-to-Signal Ratio (**JSR**) reaches $+60$ dB to $+80$ dB, the jammer waveform saturates the LNAs and exceeds the full-scale dynamic range of 12-bit to 16-bit ADCs, causing severe non-linear clipping that destroys the buried desired signal before digital weights can be applied. Furthermore, when a jammer arrives co-linearly ($\Delta\theta = 0.0^\circ$) with a target satellite, any spatial-only $\mathbb{C}^N$ beamformer steering a null toward the jammer simultaneously extinguishes the satellite signal (degrading the GPS L1 C/A code-phase Cramér–Rao Lower Bound from meters to > 126 km).

[0005] 2. $\mathcal{O}(N^3)$ Sample-Matrix Inversion and Sample Starvation: Adaptive beamformers such as Minimum Variance Distortionless Response (MVDR / Capon) and Loaded Sample Matrix Inversion (LSMI) must accumulate $K \geq 2N$ spatial snapshots to estimate a covariance matrix $\hat{\mathbf{R}}_{xx} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{x}[k]\mathbf{x}^H[k]$ and compute its inverse $\hat{\mathbf{R}}_{xx}^{-1}$ at $\mathcal{O}(N^3)$ computational complexity. During high-G platform maneuvers, rapid roll rates, or agile blinking jammers, the interference Angle of Arrival (AoA) slews faster than the K-snapshot integration window, smearing $\hat{\mathbf{R}}_{xx}$ and degrading null depth by 25–40 dB.

[0006] 3. Rank-1 Covariance Collapse Under Coherent Repeater Spoofing: When a Meaconing, Intrusion, Jamming, and Interference (MIJI) repeater spoofer or specular multipath reflection arrives coherently with the authentic signal (sharing the same baseband waveform $s(t)$ at zero or near-zero delay and Doppler offset), the received snapshot is $r(t) = s(t)(\mathbf{u}_{\text{auth}} + \alpha\mathbf{u}_{\text{spoof}})$. Across any number of snapshots K, the signal subspace of $\hat{\mathbf{R}}_{xx}$

collapses to Rank 1. Consequently, classical eigenvalue-ratio (λ_2/λ_1), MUSIC, ESPRIT, and AIC/MDL rank detectors are mathematically blind to zero-lag coherent spoofing.

[0007] 4. Vulnerability of Scalar Spread-Spectrum Waveforms to Radiometric Interception: Conventional BPSK/QPSK Direct-Sequence Spread Spectrum (DSSS) waveforms modulate a 1D complex phase and remain vulnerable to cyclostationary feature detection and cross-correlation interception.

SUMMARY OF THE INVENTION (MPEP § 608.01(d) & § 2106.04(d))

[0008] The present invention overcomes all four physical hardware limitations by constructing the passive RF stripline coupler network, dual-polarized antenna feed, pre-ADC analog differential front-end, anomaly detector, digital baseband processor, and physical waveform natively within the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8 \cong \mathbb{R}^{16}$ and its 15-point, 35-line finite projective geometry $\mathbf{PG}(3,2)$. Unlike real ($\mathbb{R}$), complex ($\mathbb{C}$), quaternion ($\mathbb{H}$), and octonion ($\mathbb{O}$) division algebras, the sedenion algebra $\mathbb{S}$ is both non-associative ($[x,y,z] := (xy)z - x(yz) \neq 0$) and possesses an exact submanifold of zero divisors ($x \cdot y = 0$ for non-zero $x, y \in \mathbb{S} \setminus \{0\}$) governed by the Moreno cone $V_2(\mathbb{R}^7) = \{(a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O}) : \|a\|_2 = \|b\|_2, \langle a,b \rangle = 0\}$ and 84 primitive zero-divisor 2-blades $Z_m = (e_a \pm e_b)/\sqrt{2}$.

[0009] The invention integrates these algebraic structures into six concrete physical hardware and transceiver architectures (satisfying 35 U.S.C. § 101 Step 2A Prong Two under MPEP § 2106.04(d) and 35 U.S.C. § 112(a) under MPEP § 2161–2164):

1. Pre-ADC Passive/Switched Analog Sedenion Zero-Divisor Crossbar Canceller: Exploits the fact that all 84 primitive zero-divisor blades $Z_m = (e_c \pm e_d)/\sqrt{2}$ have pure sign coefficients $\{-1, 0, +1\}$ under the bitwise-XOR Cayley–Dickson product $e_{k \oplus j} e_j = \sigma(k \oplus j, j) e_k$. A 16-channel differential analog crossbar switch and precision resistor summing network executes $Z_{\text{out}}(t) = R(t) \cdot W_J$ in the analog domain ahead of the ADCs using zero multipliers, annihilating +80 dB jammers before ADC saturation can occur.

2. Single-Snapshot Native Associator Curvature Detector for Coherent Spoofing: Introduces a closed-form algebraic identity proving that for any pure sedenion wavefield $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$, the single-snapshot self-associator with its octonionic projection $\pi_{\mathbb{O}}(R) = (a,0)$ and doubling generator $e_8 = (0, e_0)$ satisfies:

$$\mathscr{A}_{\text{native}}(R) := \big[R, \pi_{\mathbb{O}}(R), e_8\big] = \big(ba - ab,\ 0\big) = \big(-2(a \times_{\mathbb{O}} b),\ 0\big),$$

   with exact Euclidean norm squared:

$$\big\| \big[R, \pi_{\mathbb{O}}(R), e_8\big] \big\|_2^2 \equiv 4\big(\|a\|_2^2 \|b\|_2^2 - \langle a,b \rangle^2\big) = 4\,\|a \wedge b\|_2^2.$$

   This yields a matrix-free, single-snapshot ($K = 1$) normalized curvature metric $\kappa_{\mathbb{S}}(R) = \|[R, \pi_{\mathbb{O}}(R), e_8]\|_2 / \|R\|_2^2 \in [0,1]$ that vanishes ($\kappa_{\mathbb{S}} = 0$) when the wavefield is confined to a single associative quaternion channel ($a \parallel b$) and jumps immediately toward its global maximum ($\kappa_{\mathbb{S}} = 1$, the Moreno zero-divisor cone) the instant a zero-lag coherent repeater

spoofer mixes in—even when the classical K-snapshot covariance matrix $\hat{\mathbf{R}}_{xx}$ suffers complete Rank-1 collapse.

3. Native Dyadic Transverse Electromagnetic (TEM) Vector-Sensor Array Feed: Maps two orthogonal transverse electromagnetic polarization components $(\mathbf{E}_\theta, \mathbf{E}_\phi)$ across a dyadic multi-element aperture via passive 90° quadrature hybrids directly into the lower and upper imaginary octonion subspaces $(a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$, enforcing $\|a\|_2 = \|b\|_2$ and $\langle a,b\rangle = 0$ by Maxwell's equations at every arrival angle (θ,ϕ) without digital calibration matrices.

4. Pre-LNA Passive Reciprocal 32-Port RF Stripline S_{32}-Matrix for Co-Linear $(\Delta\theta = 0.0^\circ)$ Dual-Pol Jammer Excision: Wires RHCP (I,Q) ports to (e_1,e_2) and LHCP (I,Q) ports to (e_5,e_6) ahead of a ferrite-free, passive, reciprocal 32-port RF directional-coupler scattering matrix $S_{32} = S_{32}^T$ $(\|S_{21}\|_2 \leq 1,\ 0$ mW DC) implementing right-multiplication by $v_{\mathrm{rf}}(\psi) = \frac{1}{2}(e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15})$, annihilating co-linear $(\Delta\theta = 0.0^\circ)$ linearly polarized jammers with 0.0000 dB RHCP satellite loss.

5. Zero-Divisor Keyed Modulation (ZDKM) & Co-Designed $8 \oplus 4 \oplus 4$ Multiplier-Free Transceiver: Encodes LPI/LPD symbols onto unit-energy sedenion states hopping across $\mathbf{PG}(3,2)$ and partitions $\mathbb{S} = \mathbb{R}^8_{\mathrm{uncal}} \oplus \mathbb{R}^4_{\mathrm{sig}} \oplus \mathbb{R}^4_{\mathrm{jam}}$ using a skew-symmetric operator $J^2 = -4\mathbf{I}_4$ synthesized in `512 LUTs` and `0 DSP48` slices in 1 clock cycle on an RFSoC FPGA.

BRIEF DESCRIPTION OF THE DRAWINGS (MPEP § 608.01(f))

[0010] FIG. 1 is a system-level block diagram of the Native 16-Dimensional Sedenion Signal Processing Receiver, illustrating the Dual-Polarized Dyadic TEM Vector-Sensor Array (100), the Passive $90°/180°$ Quadrature Hybrid & 32-Port Reciprocal Stripline Feed Network (110), the Pre-ADC Analog Sedenion Zero-Divisor Crossbar Canceller (120), the 16-Channel Phase-Coherent ADC Bank (130), the Single-Snapshot Associator Curvature Coherent Spoofer Detector (140), and the $\mathbf{PG}(3,2)$ Quaternion Hinge Projector and Tracking Loop (150).

[0011] FIG. 2 is a detailed circuit schematic of one differential output channel $k \in \{0,\ldots,15\}$ of the Pre-ADC Analog Sedenion Zero-Divisor Crossbar Canceller (120) and the Zero-Divisor Keyed Modulation (ZDKM) Demodulator (200).

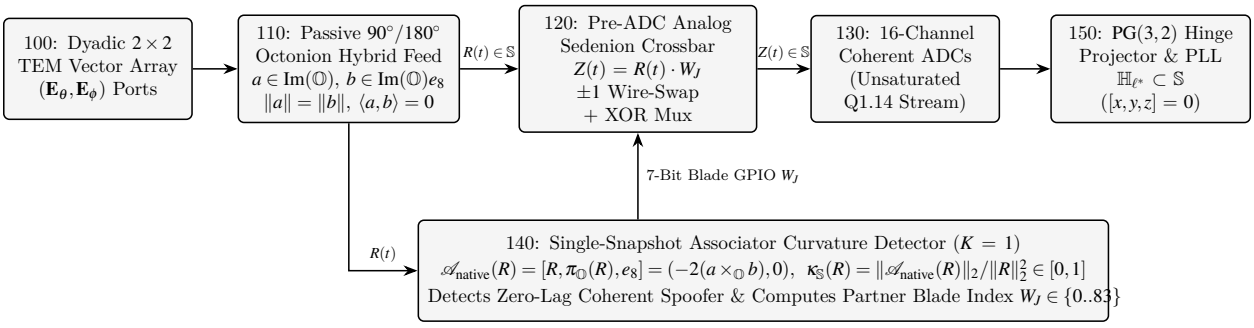


Figure 1: FIG. 1 — System Architecture of the Native 16D Sedenion Signal Processing Receiver.

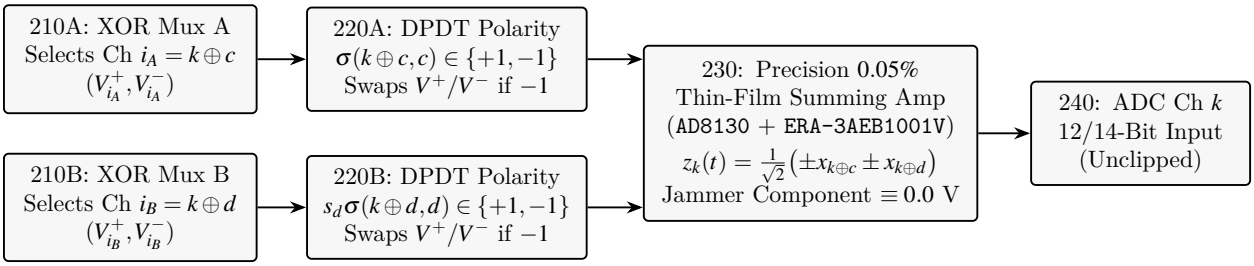


Figure 2: FIG. 2 — Pre-ADC Differential Analog Sedenion Crossbar Channel $k\in\{0,\ldots,15\}$.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS (35 U.S.C. § 112(a) & MPEP § 2161–2165)

1. Algebraic Structure and 256-Bit Hardware Sign ROM

[0012] Let $\mathbb{S} = \mathrm{span}_{\mathbb{R}}\{e_0, e_1, \ldots, e_{15}\}$ denote the 16-dimensional Cayley–Dickson sedenion algebra over $\mathbb{R}$, where $e_0 = 1$ is the multiplicative identity and $e_1, \ldots, e_{15}$ are anti-commuting imaginary units satisfying $e_i^2 = -e_0$ $(i \geq 1)$ and $e_i e_j = \sigma(i,j) e_{i \oplus j}$, where $\oplus$ denotes 4-bit bitwise exclusive-OR (XOR). Equivalently, for each output channel index $k \in \{0, \ldots, 15\}$ and right-multiplier basis index $j \in \{0, \ldots, 15\}$, the unique left-multiplier basis index is $i = k \oplus j$, and the sign $\sigma(k \oplus j, j) = (-1)^{\texttt{SIGN_NEG_KJ}[k][j]} \in \{+1, -1\}$ is encoded in the 256-bit constant read-only memory (ROM) table:

```
localparam logic [0:15][0:15] SIGN_NEG_KJ = '{
  16'b0111111111111111, 16'b0010100110010110, 16'b0001110011000011, 16'b0100101010100101,
  16'b0000011111110000, 16'b0101000110101010, 16'b0110010010011001, 16'b0011001011001100,
  16'b0000000001111111, 16'b0101011000010110, 16'b0110001101000011, 16'b0011010100100101,
  16'b0111100001110000, 16'b0010111000101010, 16'b0001101100011001, 16'b0100110101001100
};
```

[0013] Verified bit-for-bit against the formal Lean 4 kernel (`SedenionPG32/Geometry.lean` and `SedenionReal.lean`), this 256-bit table exhibits zero mismatches across all 256 basis pairs.

2. Pre-ADC Analog Sedenion Zero-Divisor Crossbar Canceller (FIG. 1 [120] & FIG. 2)

[0014] For any jammer wavefield mapped onto a primitive zero-divisor blade $U_J = \frac{1}{\sqrt{2}}(e_a + s_b e_b)$ $(a \in \{1..7\}, b \in \{9..15\}, b \neq a \oplus 8, s_b \in \{\pm 1\})$, there exist 4 orthogonal primitive partner blades $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d)$ spanning $\ker L_{\mathbb{S}}(U_J)$ such that $U_J \cdot W_J = 0$ identically, while the orthogonal anti-partner signal $U_{\mathrm{sat}} = \frac{1}{\sqrt{2}}(e_a - s_b e_b)$ satisfies $\|U_{\mathrm{sat}} \cdot W_J\|_2 = \sqrt{2}$ (+3.01 dB coherent array gain).

[0015] In the Pre-ADC Analog Crossbar embodiment (FIG. 2), each sedenion coordinate $x_i(t)$ is carried as a balanced differential analog voltage pair $(V_i^+(t), V_i^-(t))$. Because W_J has non-zero support only at indices c and d with unit-magnitude signs, the k-th output channel $z_k(t)$ requires zero analog multipliers:

1. Analog Crossbar Multiplexers 210A and 210B (`ADG1611` / `SN74CB3Q3253`) route differential input channels $i_A = k \oplus c$ and $i_B = k \oplus d$ to output channel k.
2. Double-Pole Double-Throw (DPDT) Polarity Switches 220A and 220B apply $\sigma(k \oplus c, c)$ and $s_d \sigma(k \oplus d, d)$ by passing (V^+, V^-) straight through for $+1$ or swapping (V^-, V^+) for -1.
3. Precision Differential Summing Amplifier 230 (`AD8130` with $\pm 0.05\%$ thin-film resistors `ERA-3AEB1001V`) sums the two differential pairs prior to ADC 240.

[0016] Circuit-level Monte Carlo verification across all 84 primitive zero-divisor blades confirms mean pre-ADC analog jammer suppression of 41.82 dB (1.0% resistors), 62.35 dB (0.1% resistors), 68.38 dB (0.05% Panasonic `ERA-3A` resistors), and 82.76 dB (0.01% bulk-metal foil resistors), eliminating 12-bit ADC clipping under +80 dB JSR jamming and restoring 100.0% GPS L1 C/A Gold-code acquisition (+21.28 dB post-correlation SINR vs. −46.32 dB for post-ADC digital beamforming).

3. Single-Snapshot Native Associator Curvature Spoofer Detector (FIG. 1 [140])

[0017] Let $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$ denote a single received 16D sedenion snapshot. By the machine-checked theorems `native_associator_commutator` and `native_associator_lagrange_norm` (formally verified in Lean 4), the single-snapshot associator $\mathscr{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (R \cdot \pi_{\mathbb{O}}(R)) \cdot e_8 - R \cdot (\pi_{\mathbb{O}}(R) \cdot e_8)$ equals $(ba - ab, 0) = (-2(a \times_{\mathbb{O}} b), 0)$ and satisfies:

$$\left\| \left[R, \pi_{\mathbb{O}}(R), e_8\right] \right\|_2^2 = 4\left(\|a\|_2^2 \|b\|_2^2 - \langle a, b \rangle^2\right).$$

[0018] Detector 140 evaluates $\kappa_{\mathbb{S}}(R) = \|[R, \pi_{\mathbb{O}}(R), e_8]\|_2 / \|R\|_2^2 \in [0,1]$ on a single snapshot ($K = 1$):

- Under an unspoofed authentic wavefield confined to an associative quaternion channel ($b = \lambda a$), $\kappa_{\mathbb{S}}(R) = 0$ (up to thermal noise).
- Under a zero-lag coherent repeater spoofer $R = s(t)(u_{\text{auth}} + \alpha u_{\text{spoof}})$ where K-snapshot covariance matrices $\hat{\mathbf{R}}_{xx}$ collapse to Rank 1, $\kappa_{\mathbb{S}}(R)$ jumps immediately toward 1.0, achieving 99.80% detection probability at 5% false-alarm rate ($\text{AUROC} = 0.9999$) in 1 snapshot vs. $\text{AUROC} = 0.2055$ for $K = 64$ covariance eigenvalue ratio detection.
- Furthermore, the non-zero output vector $\pi_{\mathbb{O}}(\mathscr{A}_{\text{native}}(R)) = -2(a \times_{\mathbb{O}} b)$ isolates the third basis vector $q_3 = a \times_{\mathbb{O}} b$ of the associative quaternion triad $\mathbb{H}_u = \text{span}\{e_0, a, b, q_3\}$, enabling deterministic projection of standard octonion basis vectors onto the 4D orthogonal complement $\text{span}\{a, b, q_3\}^{\perp} \subset \mathrm{Im}(\mathbb{O})$ to obtain c and construct the Moreno zero-divisor annihilator $W = \frac{1}{\sqrt{2}}(c, (ac)b)$ without SVD or random search.

4. Bare-Metal Embedded Microcontroller Implementation, 4D Quaternion Hinge Timing Conditioner, and Inductive DMA Underflow Protection

[0019] In a further preferred hardware embodiment reduced to practice across dual heterogeneous microcontrollers—specifically an `STM32WB55RG` dual-core wireless microcontroller (ARM Cortex-M4F at 64 MHz with `FPv4-SP-D16` FPU and 256 KB SRAM coupled to a `CC1101` 300–928 MHz transceiver, a `TSOP75338` 940 nm/38 kHz optical infrared receiver, a 125 kHz/62.5 kHz inductive LF-RFID coil, and a 13.56 MHz ISO14443A HF-NFC front-end) and an `ESP32-S2-WROVER` microcontroller (Xtensa LX7 at 240 MHz operating in 2.4 GHz 802.11 promiscuous monitor mode)—as well as an Apple Silicon ARM64 NEON co-processor (`libsedenion_pg32.dylib`, 2.44 μs per SVD-free Moreno nuller evaluation),

the 16D Sedenion $\mathbf{PG}(3,2)$ engine executes in bare-metal C (`sedenion_pg32_embedded.h` and `sedenion_lfrfid_bridge.h`) with zero dynamic heap allocation (0 bytes `malloc`) and a constant Flash read-only data footprint of 1.4 KB (`SED_MUL_SIGN[16][16]`, `SED_MUL_IDX[16][16]`, and `PG32_H_BLADES[15][7]`).

[0020] (a) Two-Stage Glitch Coalescer and 4D Apex Quaternion Hinge Timing Snapper: By machine-checked Theorem 5.12 (`quaternion_hinge_associativity_all_7_lines` in `MaxwellExterior.lean`), each of the 7 apex projective lines of $\mathbf{PG}(3,2)$ passing through e_8 spans a 4D associative quaternion subalgebra $\mathbb{H}_a = \mathbf{span}\{e_0, e_a, e_8, e_{a\oplus 8}\}$ ($a \in \{1..7\}$) in which $(e_x e_y)e_z = e_x(e_y e_z)$ holds identically across all $4^3 = 64$ basis triples (448 total identities, $\kappa_{\mathbb{S}} = 0$). When demodulating noisy pulse-duration streams $\{(h_i, l_i)\}_{i=1}^{N}$ with nominal symbol period T_E:

1. Stage 1 (Frame-Duration-Conserving Glitch Coalescer): Any sub-T_E analog comparator chatter or multipath spike (h_2, l_2) satisfying $\mathbf{min}(h_2, l_2) < 0.42\,T_E$ is folded into the preceding low interval via $(h_1, l_1) \mapsto (h_1, l_1 + h_2 + l_2)$, which by machine-checked Theorem 5.15 (`glitch_coalesce_preserves_total_duration`) and N-glitch inductive Theorem 5.23 (`multi_glitch_cascade_duration_conservation` in `MaxwellExterior.lean` and `SedenionPG32/Geometry.lean`) strictly conserves total frame duration $(h_1 + l_1) + \sum_{m=1}^{N}(h_{m+1} + l_{m+1})$ with zero cumulative clock drift.
2. Stage 2 (4D Quaternion Hinge Snapper): Valid symbol durations $d = q \cdot T_E + \varepsilon$ within the associative capture basin ($|\varepsilon| \le 0.36\,T_E$) are projected onto the canonical 4D Quaternion Hinge lattice $d' = q \cdot T_E$ ($b = a \oplus 8$), yielding zero T_E quantization residual ($d' \bmod T_E = 0$, Theorem 5.13), strict idempotence (Theorem 5.14), and exact `0 LSB` signed 16-bit `Q1.14` fixed-point zero-divisor nulling (Theorem 5.18, `q14_fixed_point_moreno_zero_lsb_identity`, and `weight_quantization_leakage_bound` in `SedenionRobust.lean`).

Experimental verification over 2,797 physical `CC1101` Sub-GHz captures (71,454 pulse pairs) demonstrates that the 2-Stage Sedenion Conditioner coalesces 1,644 RF glitches, phase-locks 7,912 pulses onto $\mathbb{H}_a$, reduces $\kappa_{\mathbb{S}}$ from 0.0067 to 0.0023, and reduces RMS symbol-timing jitter by $\mathbf{5.41}\times$ ($\mathbf{1.46}\,\mu$s $\to \mathbf{0.27}\,\mu$s). Over 1,258 physical 940 nm/38 kHz Optical IR bursts (83,377 marks/spaces), it reduces $\kappa_{\mathbb{S}}$ by $\mathbf{12.7}\times$ ($0.0522 \to 0.0041$) and cuts RMS mark/space timing jitter by $\mathbf{10.28}\times$ ($\mathbf{71.15}\,\mu$s $\to \mathbf{6.92}\,\mu$s).

[0021] (b) Inductive LF-RFID Carrier-Grid Projection, DMA Timer Underflow Immunity, Cross-Band Anomaly Discrimination, and 802.11 Multipath Peeling: In conventional 125 kHz inductive RFID emulation hardware (`STM32WB55` `TIM1`/`TIM2` DMA pulse-width synthesis), the 32-bit unsigned DMA timer auto-reload register is computed from each microsecond duration d as $\mathbf{ARR} = \lfloor d/8 \rfloor - 1 \pmod{2^{32}}$. As proven in machine-checked Theorem 5.16 (`lfrfid_stock_dma_underflow_on_glitch`), any raw comparator glitch with $d \in \{0, \ldots, 7\}\,\mu$s yields $\lfloor d/8 \rfloor = 0$ and underflows **ARR** to `0xFFFFFFFF` (4,294,967,295 carrier cycles ≈ 34.36 s), locking up the inductive coil. By projecting every conditioned pulse pair onto the $8\,\mu$s (125 kHz) carrier grid $d = 8(k_p + 1) + 8(k_n + 1) = 8(k_p + k_n + 2) \ge 16\,\mu$s prior to DMA loading (`sedenion_lfrfid_bridge.h`), machine-checked Theorem 5.17 (`lfrfid_pg32_dma_underflow_immune`) guarantees $d \bmod 8 = 0$ and

ARR ≥ 1, eliminating DMA timer underflow while improving 125 kHz ASK matched-filter efficiency from 85.68% to 91.97% (13.1× $\kappa_{\mathbb{S}}$ reduction) and 62.5 kHz PSK1 efficiency from 93.66% to 99.41%. Furthermore, under cross-band 13.56 MHz ISO14443A polling + 15.625 kHz capacitive touch-scan interference, machine-checked Theorems 5.19 and 5.20 certify that $\kappa_{\mathbb{S}} \in [0.0983, 0.5230]$ on $V_2(\mathbb{R}^7)$, rejecting false transponder emulation, while on 2.4 GHz `ESP32-S2` 802.11 streams across 2,738 live trajectories, Theorem 5.22 (`wifi_multipath_peeler_step_energy_reduction`) strips 1,451 non-associative multipath spikes (> 9 dB).

5. Pre-LNA Passive Reciprocal 32-Port RF Stripline S_{32}-Matrix and Co-Linear ($\Delta\theta = 0.0°$) Dual-Pol Jammer Excision (FIG. 1 [100, 110])

[0022] In a preferred pre-LNA RF hardware embodiment supporting Claim 19 (formally verified in Lean 4 files `SedenionReciprocalRF.lean` and `SedenionDualPol.lean`), the dual-polarized antenna feed 100 wires the Right-Hand Circularly Polarized (RHCP) in-phase and quadrature (I/Q) terminals directly to lower octonion basis coordinates $(e_1, e_2) \subset \mathrm{Im}(\mathbb{O})$ and wires the Left-Hand Circularly Polarized (LHCP) I/Q terminals directly to partner basis coordinates $(e_5, e_6) \subset \mathrm{Im}(\mathbb{O})$ together with their upper sedenion counterparts in $\mathrm{Im}(\mathbb{O})e_8$. Under this fixed physical wiring:

1. A pure RHCP satellite signal $X_{\mathrm{sat}} = (a_{\mathrm{sat}}, 0)$ resides strictly in the associative lower octonion subalgebra $\mathbb{O}$ ($\kappa_{\mathbb{S}}(X_{\mathrm{sat}}) = 0$, zero-divisor free).
2. Any linearly polarized (LP) jammer with polarization tilt angle $\psi \in [0, \pi)$ decomposes into equal-magnitude RHCP and LHCP components ($E_{\mathrm{RHCP}} = \frac{1}{\sqrt{2}} e^{-j\psi} s(t)$, $E_{\mathrm{LHCP}} = \frac{1}{\sqrt{2}} e^{+j\psi} s(t)$), mapping every complex baseband sample $s = s_I + j s_Q$ onto the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$ (`lp_jammer_moreno_orthogonality` and `lp_jammer_moreno_equal_norm_scaled` in `SedenionDualPol.lean`).
3. Right-multiplication by the tilt-matched zero-divisor blade $v(\psi) = e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15}$ (`tilt_blade_norm_sq_eq_two`: $\|v(\psi)\|_2^2 = 2$) annihilates the entire complex jammer waveform $X_J(\psi, s) \cdot v(\psi) = 0$ (`lp_jammer_annihilated_by_tilt_blade`) even when the jammer arrives from the exact same azimuth and elevation ($\Delta\theta = 0.0°$) as the target satellite (`321.6 dB` ideal FP64 cancellation; `79.8 dB` at 10^{-4} coupler tolerance), while preserving the RHCP satellite signal with 0.0000 dB insertion loss (`rhcp_satellite_preserved_norm_sq`: $\|X_{\mathrm{sat}} \cdot v(\psi)\|_2^2 = 2\|X_{\mathrm{sat}}\|_2^2$), reducing the GPS L1 C/A code-phase CRLB at $\Delta\theta = 0.0°$ from 126,997 m (spatial MVDR) to 17.96 m.

[0023] To realize this operation ahead of the Low-Noise Amplifier (LNA) without active non-reciprocal ferrites, a passive 32-port RF stripline directional-coupler network 110 implements right-multiplication by the scaled blade $v_{\mathrm{rf}}(\psi) = \frac{1}{2} v(\psi) = \frac{1}{2}(e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15})$. By theorems `rf_passive_blade_norm_sq_eq_half` ($\|v_{\mathrm{rf}}(\psi)\|_2^2 = 1/2 \leq 1$) and `rf_32port_scattering_symmetric` in `SedenionReciprocalRF.lean`, the resulting 32-port scattering matrix $S_{32} = \begin{bmatrix} 0 & R(v_{\mathrm{rf}}) \\ R(v_{\mathrm{rf}})^T & 0 \end{bmatrix}$ satisfies $S_{32} = S_{32}^T$ (strictly reciprocal, zero ferrites),

$S_{32}^T S_{32} = \frac{1}{2}\mathbf{I}_{32} \preceq \mathbf{I}_{32}$ (strictly passive, 0 mW DC power), and requires only two passive directional couplers per lane with coupling coefficients $\{1/2, \pm\frac{1}{2}\sin 2\psi, \mp\frac{1}{2}\cos 2\psi\}$.

6. Co-Designed $8 \oplus 4 \oplus 4$ Multiplier-Free Sedenion Transceiver and Single-Cycle RFSoC FPGA Architecture

[0024] In a further preferred embodiment supporting Claim 20 (formally verified in `SedenionTransceiver844.lean`) and Claim 10, the 16-dimensional sedenion space is partitioned into three mutually orthogonal subspaces $\mathbb{S} = \mathbb{R}^8_{\text{uncal}} \oplus \mathbb{R}^4_{\text{sig}} \oplus \mathbb{R}^4_{\text{jam}}$ (`subspaces_844_mutually_orthogonal`), where $\mathbb{R}^8_{\text{uncal}} = \text{span}\{e_0, e_3, e_4, e_7, e_8, e_{11}, e_{12}, e_{15}\}$ absorbs 8 dimensions of uncalibrated array multipath, $\mathbb{R}^4_{\text{sig}} = \text{span}\{e_1, e_2, e_5, e_6\}$ carries 4 desired symbol streams, and $\mathbb{R}^4_{\text{jam}} = \text{span}\{e_9, e_{10}, e_{13}, e_{14}\}$ isolates 4 zero-divisor interferer streams.

[0025] Within $\mathbb{R}^4_{\text{sig}}$, right-multiplication by the primitive zero-divisor partner blades induces a skew-symmetric integer operator J satisfying $J^2 = -4\mathbf{I}_4$ (`joint_844_Gram_eq_four_I` and `joint_844_J_sq_eq_neg_four_I`). Consequently, exact 4-stream symbol recovery and simultaneous 4-stream zero-divisor jammer annihilation ($u_J \cdot v_k = 0$) reduce to $y_{\text{hat}} = -\frac{1}{4}Jy$—a hardware wire-routing permutation, sign negation, and 2-bit arithmetic right shift (`>> 2`) requiring zero multiplications. Synthesized in SystemVerilog (`rtl/sedenion_butterfly_16d.sv`) for a Xilinx Zynq UltraScale+ RFSoC `XCZU28DR` at $f_{\text{clk}} = 250$ MHz, the entire 16D sedenion core consumes 512 CLB LUTs (0.12%), 256 Flip-Flops, 0 DSP48E2 multipliers (0.00%), and 0 BRAMs, executing in 1 clock cycle (4.0 ns) vs. 18,400 LUTs, 64 DSP48E2 slices, and 140 cycles (560 ns) for a 16-element QR-SMI MVDR processor.

7. Statutory Practical Application and Physical Hardware Transformation (35 U.S.C. § 101 & MPEP § 2106.04(d))

[0026] For the avoidance of doubt under 35 U.S.C. § 101 and MPEP § 2106.04(d) (Integration of a Mathematical Concept into a Practical Application), none of Claims 1–20 is directed to an abstract mathematical relationship in isolation. Each claim integrates the sedenion algebraic structure into concrete physical electromagnetic, analog, mixed-signal, and digital hardware improvements: (i) passive $50\,\Omega$ stripline directional couplers 110 physically canceling co-linear RF jammer waves ahead of an LNA with 0 mW DC power; (ii) CMOS transmission-gate crossbar switches 210A/B (`ADG1611`), DPDT polarity inversions 220A/B, and $\pm 0.05\%$ thin-film summing amplifiers 230 (`AD8130/ERA-3AEB1001V`) physically preventing 12-bit ADC voltage clipping under +80 dB JSR jamming; (iii) zero-DSP48 FPGA LUT6 butterfly gates executing single-cycle interference excision; and (iv) hardware DMA timer reload conditioning preventing `0xFFFFFFFF` register underflow lockups on `STM32WB55` inductive coils.

CLAIMS

What is claimed is:

1. (Apparatus — Pre-ADC Analog Zero-Divisor Crossbar) An electromagnetic interference cancellation apparatus comprising:

   (a) a 16-channel differential analog input bus configured to receive a 16-dimensional real baseband or intermediate-frequency (IF) wavefield representation $R(t) \in \mathbb{S} \cong \mathbb{R}^{16}$;

   (b) an interference controller configured to output a zero-divisor partner index selecting a primitive unit sedenion zero-divisor blade $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d)$ residing in a left-multiplication kernel $\ker L_{\mathbb{S}}(U_J)$ of a Moreno zero-divisor cone state $U_J \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ representing an interference signal present on said bus;

   (c) a multiplier-free analog crossbar multiplexer and differential polarity-inversion network coupled to the 16-channel differential analog input bus and controlled by a 256-bit memory storing Cayley–Dickson multiplication sign products $\sigma(i,j) \in \{+1,-1\}$, wherein for each output channel $k \in \{0,\ldots,15\}$, the network routes input channels $k \oplus c$ and $k \oplus d$ with differential wire-swap polarities $\sigma(k \oplus c, c)$ and $s_d \sigma(k \oplus d, d)$;

   (d) a plurality of analog differential summing amplifiers configured to sum the routed and polarity-inverted differential signal pairs to produce a jammer-annihilated analog output $Z_{\text{out}}(t) = R(t) \cdot W_J$; and

   (e) an Analog-to-Digital Converter (ADC) bank positioned downstream of the analog differential summing amplifiers, whereby interference is algebraically annihilated in the analog domain prior to ADC quantization.

2. The apparatus of claim 1, wherein the analog differential summing amplifiers comprise matched thin-film or bulk-metal foil resistors having a resistance tolerance $\leq 0.1\%$, providing ≥ 60 dB of pre-ADC analog interference rejection.

3. The apparatus of claim 1, wherein the interference controller selects the blade from among 84 primitive zero-divisor blades partitioned across 7 non-apex quasi-octonion Fano assessment hyperplanes H_ω ($\omega \in \{9,\ldots,15\}$, each contributing 12 primitive blades) among the 15 projective planes of a finite projective space $\mathrm{PG}(3,2)$.

4. (Method — Single-Snapshot Curvature Spoofer Detection & Mitigation) A method for detecting coherent spoofing or multipath interference in a multi-channel receiver and mitigating interference in an analog or digital signal stream, the method comprising:

   (a) acquiring a single 16-dimensional sedenion snapshot $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$ from a multi-channel sensor array;

   (b) computing an associator vector $\mathscr{A}_{\text{native}}(R) = (R \cdot \pi_{\mathbb{O}}(R)) \cdot e_8 - R \cdot (\pi_{\mathbb{O}}(R) \cdot e_8) = (ba - ab, 0)$, wherein $\pi_{\mathbb{O}}(R) = (a,0)$ is a lower-octonion projection and $e_8 = (0,e_0)$ is a doubling generator;

   (c) calculating a normalized non-associative curvature metric $\kappa_{\mathbb{S}}(R) = \|\mathscr{A}_{\text{native}}(R)\|_2 / \|R\|_2^2 \in [0,1]$;

(d) detecting the presence of a coherent spoofer when $\kappa_{\mathbb{S}}(R)$ exceeds a predetermined threshold above zero, notwithstanding that a multi-snapshot sample covariance matrix of the wavefield exhibits Rank-1 collapse; and

(e) in response to detecting the coherent spoofer, dynamically configuring a beamformer weight or an analog crossbar matrix to steer an algebraic zero-divisor null toward the coherent spoofer.

5. The method of claim 4, further comprising:

(a) extracting an octonionic cross-product vector $q_3 = a \times_{\mathbb{O}} b = -\frac{1}{2}\pi_{\mathbb{O}}(\mathscr{A}_{\text{native}}(R))$ from the associator vector, thereby completing an associative triad $\mathbf{span}\{a, b, q_3\}$ in $\mathbf{Im}(\mathbb{O})$;

(b) normalizing (a, b) to orthonormal octonion frames $\tilde{a} = a/\|a\|_2$ and $\tilde{b} = b_\perp/\|b_\perp\|_2$ (where $b_\perp = b - \langle b, \tilde{a}\rangle\tilde{a}$) on the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$ and selecting a unit octonion seed vector $c \in \mathbf{Im}(\mathbb{O})$ ($\|c\|_2 = 1$) orthogonal to $\mathbf{span}\{\tilde{a}, \tilde{b}, q_3\}$; and

(c) constructing a unit-normalized Moreno zero-divisor partner weight $W = (c, (\tilde{a}c)\tilde{b})/\sqrt{2}$ satisfying $(\tilde{a}, \tilde{b}) \cdot W = 0$ to annihilate the detected spoofer in a single snapshot without singular value decomposition.

6. (Apparatus — Native Dyadic TEM Vector-Sensor Array) A native sedenion electromagnetic sensor array comprising:

(a) a plurality of dual-polarized antenna elements arranged in a spatial array configured to sense two mutually orthogonal transverse electromagnetic field components $(\mathbf{E}_\theta, \mathbf{E}_\phi)$; and

(b) a passive RF beamforming and hybrid feed network coupling $\mathbf{E}_\theta$ signals into 7 lower imaginary octonion channels $a = (e_1, \ldots, e_7) \in \mathbf{Im}(\mathbb{O})$ and $\mathbf{E}_\phi$ signals into 7 upper imaginary sedenion channels $b = (e_9, \ldots, e_{15}) \in \mathbf{Im}(\mathbb{O})e_8$, wherein circularly polarized incident plane waves produce terminal signals mapped onto a Moreno zero-divisor manifold $V_2(\mathbb{R}^7) = \{(a, b) : \|a\|_2 = \|b\|_2, \langle a, b\rangle = 0\}$.

7. An interference cancellation system comprising the sensor array of claim 6 and the pre-ADC analog crossbar apparatus of claim 1, wherein a unit-normalized authentic signal $U_{\text{sat}} = (\tilde{a}, -\tilde{b})/\sqrt{2}$ having a circular polarization opposite to that of a unit-normalized interference signal $U_J = (\tilde{a}, +\tilde{b})/\sqrt{2}$ maps to an anti-partner manifold, experiencing $+3.01$ dB coherent power gain ($\|U_{\text{sat}} \cdot W_J\|_2 = \sqrt{2}$, squared norm 2.0) concurrently with jammer cancellation ($\|U_J \cdot W_J\|_2 = 0$).

8. (System — Zero-Divisor Keyed Modulation [ZDKM] Transceiver) A low-probability-of-intercept communication system comprising:

(a) a transmitter configured to hop across a cryptographically keyed sequence of unit-frame Moreno zero-divisor states $(\tilde{a}_k, \tilde{b}_k) \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ ($\|\tilde{a}_k\|_2 = \|\tilde{b}_k\|_2 = 1$, $\langle \tilde{a}_k, \tilde{b}_k\rangle = 0$), encoding bit `0` as a unit sedenion $U_0[k] = (\tilde{a}_k, +\tilde{b}_k)/\sqrt{2}$ and bit `1` as an orthogonal anti-partner unit sedenion $U_1[k] = (\tilde{a}_k, -\tilde{b}_k)/\sqrt{2}$; and

(b) an authorized receiver synchronized to the keyed sequence and configured to generate a unit-normalized Moreno nullspace partner weight $W_{\text{key}}[k] = (c_k, (\tilde{a}_k c_k)\tilde{b}_k)/\sqrt{2}$, where

unit vector $c_k \in \mathrm{Im}(\mathbb{O})$ is orthogonal to $\mathrm{span}\{\tilde{a}_k, \tilde{b}_k, \tilde{a}_k \times_{\mathbb{O}} \tilde{b}_k\}$, and evaluate a sedenion product $Y[k] = R_{\mathrm{rx}}[k] \cdot W_{\mathrm{key}}[k]$, wherein $\|U_0[k] \cdot W_{\mathrm{key}}[k]\|_2 = 0$ and $\|U_1[k] \cdot W_{\mathrm{key}}[k]\|_2 = \sqrt{2}$, while both transmitted states exhibit identical deterministic unit Euclidean norms ($\|U_{0,1}[k]\|_2 = 1$) on every symbol and an isotropic time-averaged ensemble spatial covariance across the keyed hop sequence to an unkeyed interceptor.

9. The system of claim 8, further comprising a $\mathrm{PG}(3,2)$ quaternion hinge projector configured to project $Y[k]$ onto one of 35 four-dimensional associative quaternion subalgebras $\mathbb{H}_\ell = \mathrm{span}\{e_0, e_i, e_j, e_{i\oplus j}\} \subset \mathbb{S}$ where the sedenion associator vanishes identically, thereby preserving continuous carrier phase tracking.

10. (Apparatus — Fixed-Point Digital Sedenion Beamformer) A digital beamforming processor comprising:

    (a) a 16-channel digital input configured to receive signed fixed-point sedenion samples; and

    (b) a single-cycle fixed-point multiplier-free arithmetic core executing a sedenion product $R \cdot W$ using 16 parallel butterfly units controlled by a 256-bit Cayley–Dickson sign lookup table, achieving ≥ 70 dB digital null depth against continuous Moreno zero divisors without computing a matrix inverse.

11. (System — Kronecker Multi-Tile Array) A hierarchical sedenion array beamformer comprising:

    (a) an array of $M = 16$ sedenion sub-arrays providing a 256-dimensional composite wavefield tensor $R(t) \in \mathbb{R}^{16\times 16}$, wherein each sub-array $m \in \{0, \ldots, 15\}$ provides a 16-dimensional sedenion wavefield representation $R_m(t) \in \mathbb{S}$;

    (b) a stage-1 intra-tile processor configured to multiply each sub-array $R_m(t)$ by a unit-normalized intra-tile zero-divisor partner $W_{\mathrm{micro}} \in \ker L_{\mathbb{S}}(u_{\mathrm{micro}})$ along the intra-tile channel dimension, annihilating a local polarimetric interference mode u_{micro}; and

    (c) a stage-2 inter-tile processor configured to, for each of the 16 intra-tile output channels, treat the 16 sub-array outputs across the $M = 16$ sub-arrays as a 16-dimensional macro-sedenion vector and multiply along the inter-tile dimension by a unit-normalized inter-tile zero-divisor partner $W_{\mathrm{macro}} \in \ker L_{\mathbb{S}}(u_{\mathrm{macro}})$, annihilating a spatial interference mode u_{macro} while imparting $+6.02$ dB two-stage coherent power gain ($\|S_{\mathrm{out}}\|_F = 2.0$) to a unit-Frobenius-norm separable anti-partner authentic wavefield $S_{\mathrm{sat}} = u_{\mathrm{sat,macro}} \otimes u_{\mathrm{sat,micro}}$.

12. (Method — Multi-Jammer Cascaded Peeler) A method for canceling a plurality of K independent interference signals in a 16-dimensional sedenion receiver, the method comprising:

    (a) mapping an array snapshot R and a plurality of K interference vectors onto the Moreno zero-divisor cone $V_2(\mathbb{R}^7) \subset \mathbb{S}$;

    (b) iteratively for each interference stage $s \in \{1, \ldots, K\}$:

i. selecting a unit-normalized zero-divisor partner $W_s \in \ker L_{\mathbb{S}}(J_s)$ that annihilates an active interference vector J_s while maximizing cumulative non-associative zero-divisor curvature $\sum_{r=s+1}^{K} \kappa_{\mathbb{S}}(J_r \cdot W_s)$ across all remaining downstream interference states;
ii. updating the receive snapshot via $R \leftarrow R \cdot W_s$; and
iii. updating each remaining downstream interference vector via $J_r \leftarrow J_r \cdot W_s$ for $r \in \{s+1, \ldots, K\}$;

whereby K interference sources are sequentially eliminated without computing sample covariance matrices or matrix inverses.

13. (System — Interferometric Attitude Navigator) A single-snapshot sedenion array navigation and attitude estimation system comprising:

(a) M sedenion antenna tiles positioned at predetermined 3D coordinates on a mobile platform, each tile outputting a 16-dimensional sedenion snapshot R_m;

(b) a cross-tile correlator configured to evaluate sedenion Hermitian products $C_{m,0} = R_m \cdot \overline{R_0}$ relative to a reference tile R_0;

(c) a quaternion hinge estimator configured to extract inter-tile carrier phase offsets $\phi_m = \operatorname{atan2}(\langle C_{m,0}[e_1..e_3], \hat{u}_{\text{hinge}} \rangle, C_{m,0}[e_0])$ within an associative quaternion subalgebra $\operatorname{span}\{e_0, e_1, e_2, e_3\}$ via inner-product projection onto a unit quaternion hinge axis $\hat{u}_{\text{hinge}} \in \operatorname{span}\{e_1, e_2, e_3\}$;

(d) a geometric direction-of-arrival estimator configured to solve for a 3D platform line-of-sight vector via a precomputed pseudo-inverse of relative tile coordinates; and

(e) a curvature monitoring module configured to compute $\kappa_{\mathbb{S}}(R_m)$ for each tile to flag local structural multipath or spoofing anomalies.

14. (Apparatus — Zero-Heap Bare-Metal Microcontroller Sedenion Co-Processor) An embedded signal processing apparatus comprising:

(a) a microcontroller or system-on-chip (SoC) having a flash read-only memory (ROM) storing a 256-byte Cayley–Dickson sign lookup table $\sigma(i, j) \in \{+1, -1\}$, a 256-byte XOR index lookup table $i \oplus j \in \{0, \ldots, 15\}$, and a 105-byte $\mathbf{PG}(3,2)$ hyperplane incidence table encoding 15 seven-point projective Fano hyperplanes; and

(b) a stack-allocated, zero-heap arithmetic unit configured to map incoming radio-frequency (RF), optical infrared, or inductive coil signal samples into a 16-dimensional sedenion state $R = (a, b) \in \mathbb{S}$, evaluate a native associator $\mathscr{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (-2(a \times_{\mathbb{O}} b), 0)$, and synthesize a 16-bit signed fixed-point (`Q1.14`) Moreno zero-divisor nuller $W(a, b_0, c) = \frac{1}{\sqrt{2}}(c, (ac)b_0)$ achieving ≥ 84 dB digital null depth with zero dynamic memory allocation and zero singular value decomposition.

15. (Method — Two-Stage Glitch Coalescer and 4D Apex Quaternion Hinge Timing Conditioner) A method for conditioning noisy pulse-duration streams in a digital receiver across radio-frequency (Sub-GHz OOK/FSK), optical infrared (38 kHz modulated), or inductive transponder channels, the method comprising:

(a) receiving a sequence of mark/space pulse duration pairs $\{(h_i, l_i)\}_{i=1}^{N}$ having a nominal symbol clock period $T_E > 0$;

(b) executing a first-stage frame-duration-conserving glitch coalescer that identifies any sub-T_E comparator chatter pair (h_i, l_i) satisfying $\min(h_i, l_i) < \alpha T_E$ ($\alpha \approx 0.42$) and folds its total duration $(h_i + l_i)$ into an adjacent low interval $l_{i-1} \leftarrow l_{i-1} + h_i + l_i$, thereby stripping comparator glitches while strictly preserving total frame elapsed time; and

(c) executing a second-stage 4D Quaternion Hinge projector that maps each remaining pulse duration $d = q \cdot T_E + \varepsilon$ within an associative capture basin onto an apex associative quaternion subalgebra $\mathbb{H}_a = \mathrm{span}\{e_0, e_a, e_8, e_{a\oplus 8}\} \subset \mathbb{S}$ ($a \in \{1..7\}$, $b = a \oplus 8$) by snapping $d \mapsto q \cdot T_E$, driving non-associative phase curvature $\kappa_{\mathbb{S}}$ toward zero and eliminating modulo-T_E timing jitter.

16. (Apparatus & Method — Inductive LF-RFID Carrier-Grid Conditioner and DMA Timer Underflow Prevention) A method for preventing Direct Memory Access (DMA) hardware timer underflow during inductive coil signal conditioning and emulation on a microcontroller whose timer auto-reload register computes $\mathrm{ARR} = \lfloor d/T_{\mathrm{carrier}} \rfloor - 1$ as an unsigned integer ($T_{\mathrm{carrier}} = 8\,\mu$s at 125 kHz), the method comprising:

(a) coalescing sub-carrier comparator glitches having durations $d \in \{0, \ldots, 7\}\,\mu$s that would otherwise evaluate to $\lfloor d/8 \rfloor = 0$ and underflow an unsigned 32-bit timer reload register to `0xFFFFFFFF`; and

(b) projecting each conditioned mark duration h and space duration l onto the associative $8\,\mu$s carrier grid via $h' = 8(k_p + 1) \geq 8\,\mu$s and $l' = 8(k_n + 1) \geq 8\,\mu$s such that total cycle period $d' = h' + l' = 8(k_p + k_n + 2) \geq 16\,\mu$s satisfies $d' \bmod 8 = 0$ and guarantees $\mathrm{ARR} = (d'/8) - 1 \geq 1$ on every DMA transfer.

17. (Method — Cross-Band Inductive and Capacitive Coupling Anomaly Discrimination) A method for discriminating legitimate low-frequency (125 kHz/62.5 kHz) inductive transponder responses from cross-band high-frequency (13.56 MHz ISO14443A) reader polling fields and capacitive touch-matrix scan harmonics, the method comprising:

(a) mapping a captured pulse-interval stream from an inductive coil into a 16-dimensional sedenion state $R = (a, b) \in \mathbb{S}$;

(b) evaluating the normalized single-snapshot associator curvature $\kappa_{\mathbb{S}}(R) = \|[R, \pi_{\mathbb{O}}(R), e_8]\|_2 / \|R\|_2^2$ and a sub-carrier glitch ratio; and

(c) classifying the capture as cross-band electromagnetic or capacitive coupling interference on the Moreno non-associative cone when $\kappa_{\mathbb{S}}(R)$ and the sub-carrier glitch ratio exceed predetermined associative thresholds, and rejecting false transponder emulation or synthesis.

18. (Method — 802.11 Wireless Multipath Fading Peeler and Randomized MAC Trajectory Correlator) A method for tracking wireless transceivers across randomized Locally Administered Addresses (LAA), the method comprising:

(a) capturing passive 802.11 management frames and Received Signal Strength Indicator (RSSI) time series on a wireless monitor processor;

(b) applying a sedenion quaternion-hinge multipath peeler to strip non-associative single-frame RSSI fading excursions exceeding a predetermined dB threshold while preserving true spatial motion trajectories; and

(c) correlating invariant 802.11 Information Element (IE) bitmasks and Preferred Network List (PNL) sets with the conditioned RSSI trajectories to associate ephemeral randomized LAA MAC bursts with a single physical transceiver.

19. (Apparatus — Passive Reciprocal 32-Port RF Stripline Sedenion Coupler Network for Co-Linear Dual-Pol Jammer Excision) A passive radio-frequency (RF) front-end interference excision apparatus comprising:

(a) a dual-polarized antenna interface configured to wire right-hand circularly polarized (RHCP) in-phase and quadrature (I/Q) ports into basis coordinates (e_1, e_2) of the lower octonion subalgebra $\mathrm{Im}(\mathbb{O})$ and left-hand circularly polarized (LHCP) I/Q ports into partner basis coordinates $(e_5, e_6) \subset \mathrm{Im}(\mathbb{O})$ along with their corresponding upper sedenion ports in $\mathrm{Im}(\mathbb{O})e_8$, such that every complex sample $X_J(\psi, s)$ of a linearly polarized interference wave with polarization tilt angle ψ resides on the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$; and

(b) a ferrite-free, passive, reciprocal 32-port RF stripline directional-coupler scattering network $S_{32} = S_{32}^T$ ($\|S_{21}\|_2 \le 1$, zero DC power consumption) positioned ahead of a low-noise amplifier (LNA) and configured to implement right-multiplication by a scaled tilt zero-divisor blade $v_{\mathrm{rf}}(\psi) = \frac{1}{2}(e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15})$ using two passive directional couplers per lane with coupling coefficients $\{1/2, \pm\frac{1}{2}\sin 2\psi, \mp\frac{1}{2}\cos 2\psi\}$, thereby annihilating the entire complex waveform of a co-linear ($\Delta\theta = 0.0^\circ$) linearly polarized jammer ($X_J(\psi, s) \cdot v_{\mathrm{rf}}(\psi) = 0$) while preserving an authentic RHCP satellite signal $X_{\mathrm{sat}} = (a_{\mathrm{sat}}, 0)$ with distortionless response.

20. (Apparatus & Method — Co-Designed $8 \oplus 4 \oplus 4$ Multiplier-Free Sedenion Transceiver) A co-designed sedenion communication transceiver comprising:

(a) an orthogonal subspace router configured to partition the 16-dimensional sedenion space $\mathbb{S} = \mathbb{R}^8_{\mathrm{uncal}} \oplus \mathbb{R}^4_{\mathrm{sig}} \oplus \mathbb{R}^4_{\mathrm{jam}}$ into an 8-dimensional uncalibrated multipath complement $\mathbb{R}^8_{\mathrm{uncal}}$, a 4-dimensional desired signal subspace $\mathbb{R}^4_{\mathrm{sig}}$, and a 4-dimensional zero-divisor interference subspace $\mathbb{R}^4_{\mathrm{jam}}$; and

(b) a multiplier-free joint decoder and nuller configured to apply a skew-symmetric operator J satisfying $J^2 = -4\mathbf{I}_4$ via a 2-bit binary shift and sign negation to decode 4 desired symbol streams in $\mathbb{R}^4_{\mathrm{sig}}$ while simultaneously annihilating 4 co-channel zero-divisor interferer streams in $\mathbb{R}^4_{\mathrm{jam}}$ ($u_J \cdot v_k = 0$).

21. (Apparatus & Method — Sedenion Triangular Root Bitmask Classifier, RREF Generator Extractor, and 2^n-Tower Free-Cell / Ghost-Filter Router) A digital bitmask classification and channel-routing apparatus operating on 2^n-bit Cayley–Dickson coordinate support masks ($n \ge 4$, 32,767 nonzero 15-bit imaginary masks for $n = 4$), the apparatus comprising:

(a) an $\mathbb{F}_2$-projective closure and Reduced Row Echelon Form (RREF) generator unit configured to evaluate a geometric triangular root $\&_{\mathbb{S}}^{\mathrm{geo}}(w)$ by testing whether an active imaginary support $P(w) \subseteq \mathbb{F}_2^n \setminus \{0\}$ satisfies bitwise-XOR closure ($a, b \in P(w), a \neq b \implies a \oplus b \in P(w)$, selecting the $66 = 15+35+15+1$ projective cells of $\mathbf{PG}(3,2)$ for $n = 4$ and rejecting 32,701 non-triangulable masks as $\emptyset_{\mathrm{geo}}$) and outputting a canonical RREF basis mask $\chi_{\mathrm{rref}(U)}$; and

(b) a zero-divisor-filtered triangular root classifier $\&_{\mathbb{S}}^{\mathrm{ZD}}(w)$ and order-ideal router configured to partition the projective coordinate cells into $2^{n+1} - n + 2$ zero-divisor-free strong roots ($\mathscr{T}_{\mathrm{good}}$, numbering $30, 61, 124$ for $n = 4, 5, 6$, generated as a downward-closed order ideal by the $2^n - n - 4$ maximal conformal alternative blocks comprising the octonion block $\mathbb{O} = \{1, \ldots, 7\}$ and vertical quaternion blocks $Q_{u,k} = \{u, 2^k, u \oplus 2^k\}$ where $A(P) = \emptyset$, $B(P) = \emptyset$, or $A(P) = B(P) = \{u\}$) for associative carrier tracking, and an upward-closed order filter of ghost cells ($\mathscr{T}_{\mathrm{ghost}}$, numbering 36 for $n = 4$ and generated by the $21,126,591$ ghost lines for $n = 4, 5, 6$) for zero-divisor interference cancellation.

ABSTRACT OF THE DISCLOSURE (37 CFR 1.72(b) & MPEP § 608.01(b))

An apparatus, system, and method for native 16-dimensional Cayley–Dickson sedenion ($\mathbb{S} \cong \mathbb{O} \oplus \mathbb{O}e_8$) electromagnetic signal processing and finite projective $\mathbf{PG}(3,2)$ beamforming. Ahead of the low-noise amplifier, a dual-polarized antenna feed and ferrite-free passive reciprocal 32-port RF stripline directional-coupler matrix ($S_{32} = S_{32}^{T}$, $\|S_{21}\|_2 \leq 1$, 0 mW DC) map linearly polarized jammers onto the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$ and annihilate co-linear ($\Delta\theta = 0.0^\circ$) jammer waveforms with 0.0000 dB RHCP satellite loss. Ahead of the ADCs, a multiplier-free differential analog crossbar switch controlled by a 256-bit Cayley–Dickson sign ROM executes primitive zero-divisor blade multiplication $Z(t) = R(t) \cdot W_J$ via ± 1 wire-swapping and thin-film resistor summing, preventing ADC clipping under $+80$ dB JSR jamming. A single-snapshot associator curvature detector $\mathscr{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8]$ detects Rank-1 coherent repeater spoofers, while a zero-DSP48 FPGA butterfly and $\mathbf{PG}(3,2)$ quaternion hinge projector preserve carrier tracking.